进度汇报

2020.6.5

胡敬玉

Xie Y, Lu H, Zhang J, et al. Deep Segmentation-Emendation Model for Gland Instance Segmentation， in MICCAI, 2019: 469-477.

基本信息 The basic information

综合介绍 General Introduction

夏勇，男，西北工业大学教授、博导，1997年考入西北工业大学教育实验学院，分别于2001、2004和2007年从西北工业大学计算机学院获得学士、硕士和博士学位（获中国计算机学会优秀博士论文奖）；2007年1月加入悉尼大学计算机学院（原信息技术学院）生物医学与多媒体技术（BMIT）实验室，在冯大淦院士指导下进行博士后研究，2013年入选国家"青年海外人才"计划，并于同底回到西北工业大学计算机学院，开展医学影像大数据分析、计算机辅助诊断和深度学习等领域的教学和科研工作，现为空天地海一体化大数据应用技术国家工程实验室成员；主持国家自然基金面上项目两项，近三年在IEEE会刊、MedIA、IJCAI和MICCAI等本领域顶级期刊和会议发表论文近40篇，指导学生获得了ISBI 2019"急性白血病恶性B-淋巴母细胞分类竞赛"第一名，还先后在另外三项国际医学影像分析竞赛中排名第一；现为中国计算机学会青工委委员、中国图象图形学学会视觉大数据专委会常委、中国生物医学工程学会青委会委员和中国抗癌协会肿瘤影像专业委员会人工智能学组副组长、MICS指导委员会委员和VALSE执行领域主席之一；先后担任ACM MM2018、ISBI 2017、MICCAI 2019和多个MICCAI Workshop的TCP、Session Chair或Area Chair。

问题（大肠腺癌）

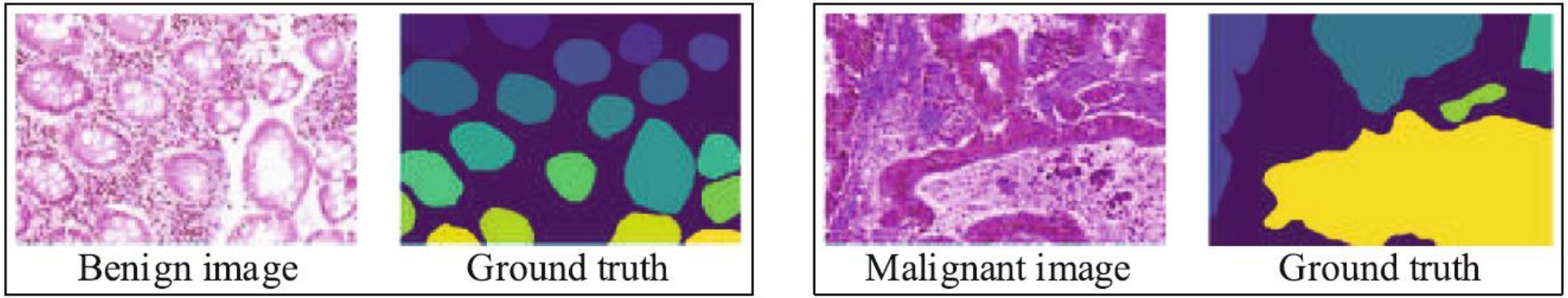

Fig. 1. Two examples of gland instance segmentation in benign and malignant cases. Individual glands are denoted by different colors. (Color figure online)

- 神经网络有效在多个医学图像分割任务上是有效的
- 将卷积神经网络应用与腺体实例分割

- 集中于优化模型结构和损失函数来提高分割性能

Segmentation-Emendation

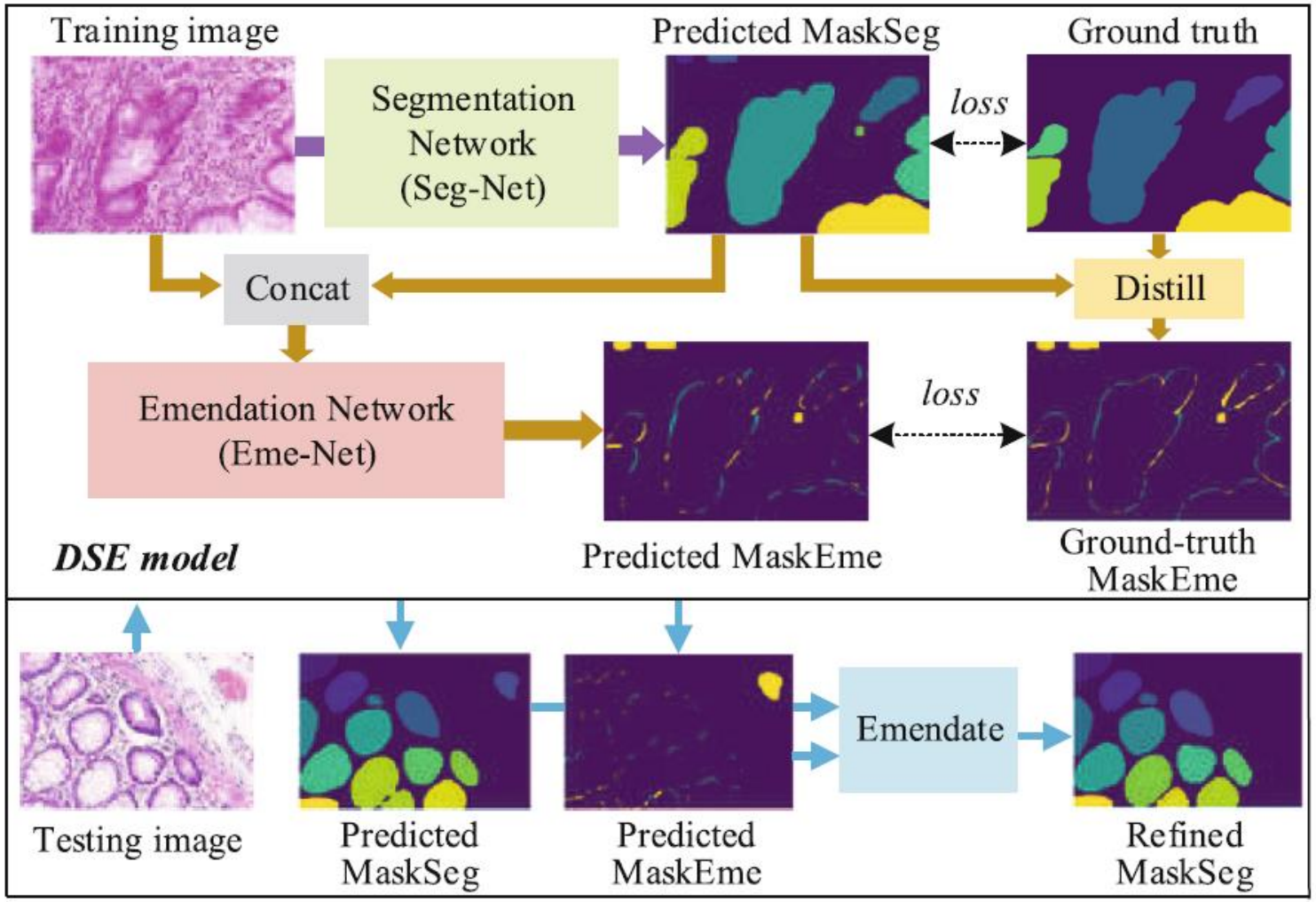

Fig. 2. The technical pipeline of our proposed DSE model.

Segmentation Network

Segmentation Network: We employ the advanced semantic segmentation network Deeplabv3+ [8] pre-trained on MS-COCO [9] and PASCAL VOC 2012 datasets [10], as the backbone of Seg-Net. To adapt Deeplabv3+ to our segmentation task, we remove its last convolutional layer, and then add a new convolutional layer with two output channels and a softmax layer for prediction. The weights attached to new layers are randomly initialized. We optimize the Seg-Net by minimizing the cross-entropy loss.

Emendation Network

quality of Seg-Net and emendate the wrong predictions. We define the ground-truth MaskEme as the pixel-wise consistency between the binary MaskSeg predicted by Seg-Net and its ground truth. Formally, the value of the i-th pixel in the ground-truth MaskEme, $\hat{E}_i$, can be denoted by:

$$\hat{E}_i = \begin{cases} 0 & if\ S_i = \hat{S}_i \\ 1 & if\ S_i < \hat{S}_i \\ 2 & if\ S_i > \hat{S}_i \end{cases} \tag{1}$$

where S_i and $\hat{S}_i$ represent the value of the i-th pixel in the predicted MaskSeg and segmentation ground truth, respectively. If $\hat{E}_i = 0$, it means that the prediction S_i is accurate and no emendation is needed. Otherwise, the prediction is either under-segmentation (i.e. $\hat{E}_i = 1$) or over-segmentation (i.e. $\hat{E}_i = 2$), and need be emendated in the inference stage.

Training Emendation Network

$$\mathcal{L} = \mathcal{L}_{Dice} + \mathcal{L}_{ce} = \frac{1}{C}\sum_{c=1}^{C}\omega_c\left[1 - \frac{2\sum_{i=1}^{V}\hat{E}_i^c E_i^c}{\sum_{i=1}^{V}(\hat{E}_i^c + E_i^c) + \varepsilon}\right] - \frac{1}{V}\sum_{v=1}^{V}\sum_{c=1}^{C}\omega_c \hat{E}_i^c log E_i^c \tag{2}$$

where C is the number of categories, V denotes the number of pixels, E_i^c indicates the prediction of i-th pixel belonging to the class c, $\hat{E}_i^c$ represents the target value of i-th pixel in the MaskEme, ε is a smoothing factor, and ω_c is the weight of the class c. The weight ω_c can be formulated as $\omega_c = log\frac{V}{V^c}$, where V^c denotes the number of pixels belonging to the class c.

Inference Process

$$S_i^{'} = \begin{cases} S_i & if\ E_i = 0 \\ 1 & if\ E_i = 1 \\ 0 & if\ E_i = 2 \end{cases} \tag{3}$$

where S_i and E_i represent the value of i-th pixel of the predicted MaskSeg and MaskEme, respectively. If $E_i = 1$, we emendate the prediction S_i as the gland class $\{1\}$. If $E_i = 2$, we emendate the prediction S_i as the background class $\{0\}$.

实验结果

Table 1. Comparison with five methods on the GlaS Challenge dataset. S and R denote score and rank, respectively

Methods	Object F1				Object Dice				Object Hausdorff				Rank sum
	Part A		Part B		Part A		Part B		Part A		Part B		
	S	R	S	R	S	R	S	R	S	R	S	R	
DSE model (Ours)	**0.926**	**1**	**0.862**	**1**	**0.927**	**1**	**0.871**	**1**	**31.209**	**1**	**80.509**	**1**	**6**
MILD-Net [2]	0.914	3	0.844	2	0.913	2	0.836	3	41.540	2	105.890	2	14
SPL-Net [4]	0.924	2	0.844	2	0.902	4	0.840	2	49.881	5	106.075	3	18
DMCN [6]	0.893	5	0.843	4	0.908	3	0.833	4	44.129	3	116.821	4	23
DCAN [5]	0.912	4	0.716	5	0.897	5	0.781	6	45.418	4	160.347	6	30
MPCNN [11]	0.891	6	0.703	6	0.882	6	0.786	5	57.413	6	145.575	5	34

Table 2. Comparison with two methods on the CRAG dataset.

Methods	Object F1		Object Dice		Object Hausdorff		Rank sum
	S	R	S	R	S	R	
DSE model (Ours)	**0.835**	**1**	**0.889**	**1**	**120.127**	**1**	**3**
MILD-Net [2]	0.825	2	0.875	2	160.140	2	6
DCAN [5]	0.736	3	0.794	3	218.76	3	9

实验结果

Table 3. Performance of the Seg-Net and our DSE model on two datasets.

Datasets	Method	Object F1	Object Dice	Object Hausdorff
GlaS Part A	Seg-Net	0.888	0.908	41.009
	DSE model	**0.926**	**0.927**	**31.209**
GlaS Part B	Seg-Net	0.827	0.855	89.090
	DSE model	**0.862**	**0.871**	**80.509**
CRAG	Seg-Net	0.774	0.850	155.430
	DSE model	**0.835**	**0.889**	**120.127**

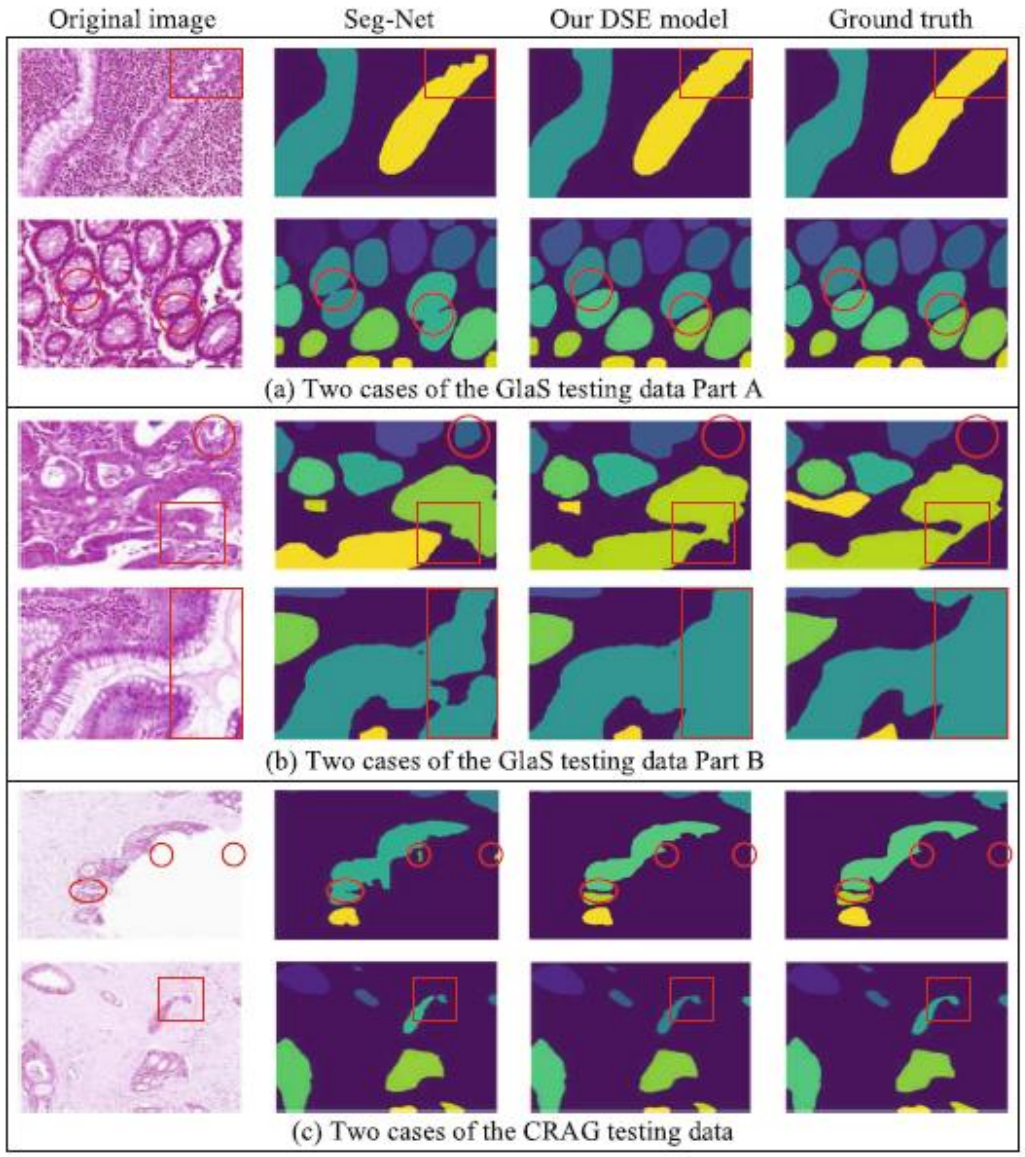

Fig. 3. Visualization of gland segmentation results of six cases selected from the GlaS and CRAG datasets. Multiple gland instances are highlighted by different colors. (Color figure online)

Thank You!